Lecture 4

Schur's lemma, orthogonality, and the group algebra

Objectives. *A single, almost trivial-looking observation about intertwiners — Schur's lemma — turns out to control the entire structure of finite group representations. From it we derive the orthogonality of matrix elements, and, via the regular representation and the group algebra, the two great counting theorems: the squared dimensions of the irreducible representations sum to the group order, and the number of irreducible representations equals the number of conjugacy classes. By the end of the lecture, the representation theory of a finite group is reduced to a finite, sharply constrained bookkeeping problem.*

4.1 Schur's lemma

Recall from Lecture 3 that an *intertwiner* between representations (ρ_V, V) and (ρ_W, W) of G is a linear map $T\colon V \to W$ with

$$T\,\rho_V(g) = \rho_W(g)\,T \qquad \text{for all } g \in G;$$

there we required T invertible and called the representations equivalent. Now we drop invertibility and study the vector space of *all* intertwiners, denoted $\mathrm{Hom}_G(V,W)$. The decisive observation is one line long:

Lemma 4.1. *If $T \in \mathrm{Hom}_G(V,W)$, then $\ker T$ is an invariant subspace of V and $\operatorname{im} T$ is an invariant subspace of W.*

Proof. If $Tv = 0$ then $T\rho_V(g)v = \rho_W(g)Tv = 0$, so $\rho_V(g)v \in \ker T$. And $\rho_W(g)\,Tv = T\rho_V(g)v \in \operatorname{im} T$. ∎

For *irreducible* representations, invariant subspaces are scarce, and the lemma snaps shut into a powerful dichotomy.

Theorem 4.2 (Schur's lemma, first form)**.** *Let V and W be irreducible representations of G and $T \in \mathrm{Hom}_G(V,W)$. Then either $T = 0$ or T is an isomorphism (so $V \cong W$). In particular, between* inequivalent *irreducible representations the only intertwiner is zero.*

Proof. Suppose $T \neq 0$. By Lemma 4.1, $\ker T$ is an invariant subspace of the irreducible V, and $\ker T \neq V$ since $T \neq 0$; hence $\ker T = \{0\}$ and T is injective. Likewise $\operatorname{im} T$ is a nonzero invariant subspace of the irreducible W, hence $\operatorname{im} T = W$: T is surjective. ∎

Theorem 4.3 (Schur's lemma, second form)**.** *Let (ρ, V) be a* finite-dimensional complex *irreducible representation, and let $T \in \mathrm{Hom}_G(V,V)$, i.e. $T\rho(g) = \rho(g)T$ for all g. Then $T = \lambda\ \mathrm{id}_V$*

for some $\lambda \in \mathbb{C}$:

$$\operatorname{Hom}_G(V,V) = \mathbb{C}\ \mathrm{id}_V\,.$$

Proof. Because $\mathbb{C}$ is algebraically closed, the characteristic polynomial of T has a root: T has an eigenvalue $\lambda \in \mathbb{C}$. Then $T - \lambda\ \mathrm{id}_V$ is again an intertwiner (sums and scalar multiples of intertwiners intertwine, and id_V does trivially), and its kernel — the λ-eigenspace — is nonzero. By Theorem 4.2 applied to $T - \lambda\ \mathrm{id}_V$, a non-invertible intertwiner from V to V must vanish: $T = \lambda\ \mathrm{id}_V$. ∎

Corollary 4.4. *For complex irreducible representations V, W of a group G,*

$$\dim \operatorname{Hom}_G(V,W) = \begin{cases} 1, & V \cong W, \\ 0, & V \ncong W. \end{cases}$$

Proof. If $V \ncong W$ this is the first form. If $V \cong W$, fix an isomorphism $T_0 \in \operatorname{Hom}_G(V,W)$; for any $T \in \operatorname{Hom}_G(V,W)$ the composite $T_0^{-1}T \in \operatorname{Hom}_G(V,V)$ is a scalar λ id by the second form, so $T = \lambda T_0$. ∎

Corollary 4.5 (Abelian groups)**.** *Every complex irreducible representation of an abelian group is one-dimensional.*

Proof. Let (ρ, V) be irreducible and G abelian. For fixed $h \in G$ the operator $\rho(h)$ commutes with every $\rho(g)$ — that is, $\rho(h) \in \operatorname{Hom}_G(V,V)$ — so $\rho(h) = \lambda_h\ \mathrm{id}_V$ by the second form. But if every group element acts as a scalar, then *every* subspace of V is invariant, and irreducibility forces $\dim V = 1$. ∎

This explains, from first principles, what Lecture 3 found by hand: the rotation representation of C_3 *had* to split into one-dimensional pieces over $\mathbb{C}$. The one-dimensional representations $\chi_k(r^m) = e^{2\pi \mathrm{i} km/n}$ of C_n are therefore the only candidates for its irreducible representations; that they form the *complete* list will drop out of Section 4.3.

Remark 4.6 (Complex is essential)**.** The second form fails over $\mathbb{R}$, where eigenvalues need not exist. Let $C_4 = \langle r \rangle$ act irreducibly on $\mathbb{R}^2$ by the quarter-turn $J = R(\pi/2) = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Every matrix $a\nVdash + bJ$ commutes with the action: the commutant is two-dimensional — it is a copy of $\mathbb{C}$ hiding inside the real 2×2 matrices, with J playing i. Over $\mathbb{C}$, of course, the representation splits into $\chi_1 \oplus \chi_{-1}$ and Schur is restored. Throughout this course "irreducible" means irreducible over $\mathbb{C}$ unless flagged otherwise.

Remark 4.7 (The physics of Schur's lemma)**.** Read physically, the second form says: *an operator that commutes with an irreducible symmetry action is a multiple of the identity*. If the Hamiltonian of a system commutes with all the $\rho(g)$ — the defining property of a symmetry, as Lecture 7 will set up — then on an irreducible representation occurring with multiplicity one the Hamiltonian is forced to be a scalar: one energy for the whole representation. More generally, on an isotypic component $V^{(i)} \otimes \mathbb{C}^{m_i}$ the symmetry acts as $\rho^{(i)}(g) \otimes \nVdash$, and its commuting operators have the form $\nVdash \otimes F_i$. They may therefore mix the m_i equivalent copies; after diagonalising F_i, each of its eigenvalues still has the $\dim V^{(i)}$-fold multiplet forced by symmetry. More precisely, if an eigenvalue of F_i has multiplicity q, its total degeneracy on the isotypic component is $q \dim V^{(i)}$; only the factor $\dim V^{(i)}$ is forced by this complex irreducible type. Irreducible representations are thus the natural candidates for the degenerate multiplets of quantum mechanics, and Schur's lemma is the reason. The same mechanism, applied to operators that intertwine *different* irreducible blocks, will give the Wigner–Eckart theorem and the selection rules of Lecture 7.

4.2 Orthogonality of matrix elements

Schur's lemma is a statement about operators. Combined with the averaging trick of Lecture 3, it becomes a statement about *numbers* — the matrix elements of the irreducible representations — and these numbers turn out to be orthogonal in a precise sense.

Fix once and for all a complete list $\rho^{(1)}, \rho^{(2)}, \dots$ of pairwise inequivalent irreducible representations of the finite group G, on spaces $V^{(i)}$ of dimensions n_i, and choose bases making the matrices $D^{(i)}(g)$ *unitary* (always possible, Lecture 3). The completeness of the list — that there are finitely many and that we have them all — is established in the next section; nothing in the present one depends on it.

We regard each matrix element as a complex-valued *function on the group*, $g \mapsto D^{(i)}_{\mu\nu}(g)$, and we equip the $|G|$-dimensional space of all functions $\psi\colon G \to \mathbb{C}$ with the inner product

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)}\, \psi(g),$$

conjugate-linear in the first argument as always.

The bridge from Schur to orthogonality is the following averaged intertwiner.

Lemma 4.8. *For any linear map $A\colon V^{(j)} \to V^{(i)}$, the average*

$$T_A = \frac{1}{|G|} \sum_{g \in G} \rho^{(i)}(g)\, A\, \rho^{(j)}(g)^{-1}$$

is an intertwiner: $T_A \in \mathrm{Hom}_G\big(V^{(j)}, V^{(i)}\big)$.

Proof. For $h \in G$,

$$\rho^{(i)}(h)\, T_A\, \rho^{(j)}(h)^{-1} = \frac{1}{|G|} \sum_{g \in G} \rho^{(i)}(hg)\, A\, \rho^{(j)}(hg)^{-1} = T_A,$$

since $g \mapsto hg$ merely reorders the sum (Latin-square property, Lecture 1). Multiply on the right by $\rho^{(j)}(h)$. ∎

Theorem 4.9 (Great orthogonality theorem)**.** *With unitary matrices as above,*

$$\Big\langle D^{(i)}_{\mu\nu},\, D^{(j)}_{\alpha\beta} \Big\rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{D^{(i)}_{\mu\nu}(g)}\, D^{(j)}_{\alpha\beta}(g) = \frac{1}{n_i}\, \delta_{ij}\, \delta_{\mu\alpha}\, \delta_{\nu\beta}.$$

Equivalently: the rescaled matrix elements $\sqrt{n_i}\, D^{(i)}_{\mu\nu}$ form an orthonormal set *of functions on G.*

Proof. Apply Lemma 4.8 with $A = E^{\nu\beta}$, the matrix with a single 1 in row ν, column β, and zeros elsewhere. In matrix form, the (μ, α) entry of T_A is

$$\big(T_{E^{\nu\beta}}\big)_{\mu\alpha} = \frac{1}{|G|} \sum_{g} D^{(i)}_{\mu\nu}(g)\, D^{(j)}_{\beta\alpha}(g^{-1}) = \frac{1}{|G|} \sum_{g} D^{(i)}_{\mu\nu}(g)\, \overline{D^{(j)}_{\alpha\beta}(g)},$$

where unitarity was used in the last step: $D^{(j)}(g^{-1}) = D^{(j)}(g)^{\dagger}$, so $D^{(j)}_{\beta\alpha}(g^{-1}) = \overline{D^{(j)}_{\alpha\beta}(g)}$.

Case $i \neq j$. The representations are inequivalent, so Theorem 4.2 gives $T_A = 0$ for every A: each entry above vanishes, which is the assertion with $\delta_{ij} = 0$.

Case $i = j$. By Theorem 4.3, $T_A = \lambda(A)\,\mathrm{id}$, and taking traces fixes the scalar:

$$\lambda(A)\,n_i = \operatorname{tr} T_A = \frac{1}{|G|}\sum_g \operatorname{tr}\bigl(\rho^{(i)}(g) A \rho^{(i)}(g)^{-1}\bigr) = \operatorname{tr} A,$$

so $\lambda(E^{\nu\beta}) = \delta_{\nu\beta}/n_i$. The (μ,α) entry of $\lambda\,\mathrm{id}$ is $\delta_{\nu\beta}\,\delta_{\mu\alpha}/n_i$. Combining the two cases and taking the complex conjugate of the whole identity (harmless, as the right-hand side is real) gives the stated formula. ∎

Corollary 4.10. $\sum_i n_i^2 \leq |G|$*: a finite group has finitely many inequivalent irreducible representations, and their squared dimensions sum to at most the group order.*

Proof. The functions $\sqrt{n_i}\,D^{(i)}_{\mu\nu}$, of which there are $\sum_i n_i^2$, are orthonormal, hence linearly independent, in the $|G|$-dimensional space of functions on G. ∎

The inequality will sharpen to an equality in Section 4.3. First, a hands-on check that the theorem says what it says.

Example 4.11 (Orthogonality in D_3, numerically)**.** The group D_3 has, among its irreducible representations, the trivial representation $\mathbf{1}$, the orientation representation ν (both one-dimensional), and the two-dimensional defining representation with the unitary (indeed real orthogonal) matrices of Lecture 3, namely $D(r^k) = R(2\pi k/3)$ and $D(r^k s) = R(2\pi k/3)\,\mathrm{diag}(-1,1)$. Writing $c_k = \cos\frac{2\pi k}{3}$, $s_k = \sin\frac{2\pi k}{3}$, the matrix-element functions take the values

g	e	r	r^2	s	rs	r^2s
$\nu(g)$	1	1	1	-1	-1	-1
$D_{11}(g)$	1	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$\frac{1}{2}$	$\frac{1}{2}$
$D_{12}(g)$	0	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$

(the reflection columns use $D(r^k s)_{11} = -c_k$, $D(r^k s)_{12} = -s_k$). Then, entry by entry:

$$\langle D_{11}, D_{11}\rangle_G = \tfrac{1}{6}\bigl(1 + \tfrac14 + \tfrac14 + 1 + \tfrac14 + \tfrac14\bigr) = \tfrac12 = \tfrac{1}{n_i}, \qquad \langle D_{12}, D_{12}\rangle_G = \tfrac16 \cdot 3 = \tfrac12,$$

while the cross terms cancel in pairs:

$$\langle D_{11}, D_{12}\rangle_G = 0, \qquad \langle \mathbf{1}, \nu\rangle_G = \tfrac16(1+1+1-1-1-1) = 0, \qquad \langle \nu, D_{11}\rangle_G = 0.$$

Six functions $(1+1+4)$, pairwise orthogonal, in a six-dimensional function space: for D_3 the matrix elements visibly fill the whole space — a foretaste of the completeness theorem below.

4.3 The group algebra and the regular representation

To convert the inequality of Corollary 4.10 into an equality, we deploy one special representation that contains everything: the group acting on itself.

Definition 4.12 (Group algebra)**.** The *group algebra* $\mathbb{C}[G]$ is the $|G|$-dimensional vector space with one basis vector e_g for each $g \in G$, equipped with the bilinear multiplication extending the group law:

$$\Big(\sum_g a_g\, e_g\Big)\Big(\sum_h b_h\, e_h\Big) = \sum_{g,h} a_g b_h\, e_{gh}.$$

It is an associative algebra with unit e_e, in which G sits as the basis. The *regular representation* of G is left multiplication on this space:

$$\rho^{\mathrm{reg}}(g)\, e_h = e_{gh},$$

the permutation representation (Lecture 3) of the action of G on itself by left translation; its dimension is $|G|$.

Every representation ρ of the group extends to the algebra by linearity, $\rho(\sum a_g e_g) := \sum a_g \rho(g)$, a fact we will use silently below and seriously in the starred subsection.

Two short lemmas unlock the structure of the regular representation. The first says that an intertwiner *out of* $\mathbb{C}[G]$ is freely specified by one vector; the second says that intertwiner dimensions count multiplicities.

Lemma 4.13. *For any representation (ρ_W, W) of G, the evaluation map*

$$\mathrm{Hom}_G\big(\mathbb{C}[G],\, W\big) \longrightarrow W, \qquad T \longmapsto T(e_e),$$

is a linear isomorphism. In particular $\dim \mathrm{Hom}_G(\mathbb{C}[G], W) = \dim W$.

Proof. An intertwiner is determined by its value at e_e:

$$T(e_g) = T\big(\rho^{\mathrm{reg}}(g)e_e\big) = \rho_W(g)\, T(e_e).$$

Conversely, given any $w \in W$, define $T(e_g) := \rho_W(g)w$ and extend linearly; then $T(\rho^{\mathrm{reg}}(h)e_g) = T(e_{hg}) = \rho_W(hg)w = \rho_W(h)T(e_g)$, so T intertwines. The two constructions are mutually inverse and linear in w. ∎

Lemma 4.14 (Multiplicities count intertwiners)**.** *Let $V \cong \bigoplus_i m_i V^{(i)}$ be a decomposition of a representation into pairwise inequivalent irreducibles $V^{(i)}$ with multiplicities m_i. Then for each j,*

$$\dim \mathrm{Hom}_G\big(V,\, V^{(j)}\big) = m_j = \dim \mathrm{Hom}_G\big(V^{(j)},\, V\big).$$

Proof. Hom_G is additive in direct sums in either slot: an intertwiner from a direct sum is the collection of intertwiners from its summands, and likewise into. Hence

$$\dim \mathrm{Hom}_G\Big(\bigoplus_i m_i V^{(i)},\, V^{(j)}\Big) = \sum_i m_i \dim \mathrm{Hom}_G\big(V^{(i)}, V^{(j)}\big) = \sum_i m_i\, \delta_{ij} = m_j,$$

by Corollary 4.4; the second equality is identical. ∎

Note that Lemma 4.14 settles a debt from Lecture 3: the multiplicities in a complete decomposition are the dimensions of intertwiner spaces, which are defined without reference to any decomposition — so *the multiplicities are unique*, whatever basis or splitting is used to compute them.

Theorem 4.15 (Decomposition of the regular representation)**.** *Every irreducible representation $V^{(i)}$ of the finite group G occurs in the regular representation, with multiplicity equal to its dimension:*

$$\mathbb{C}[G] \;\cong\; \bigoplus_i n_i V^{(i)}, \qquad n_i = \dim V^{(i)},$$

the sum running over all inequivalent irreducible representations. Consequently

$$\boxed{\sum_i n_i^2 = |G|}$$

Proof. By Maschke's theorem, $\mathbb{C}[G] \cong \bigoplus_i m_i V^{(i)}$ for some multiplicities $m_i \geq 0$ (a priori, some irreducibles might be absent). By Lemmas 4.13 and 4.14,

$$m_i = \dim \mathrm{Hom}_G\big(\mathbb{C}[G], V^{(i)}\big) = \dim V^{(i)} = n_i > 0 :$$

every irreducible occurs, n_i times. Comparing dimensions, $|G| = \dim \mathbb{C}[G] = \sum_i n_i \cdot n_i$. ∎

The regular representation is thus the mother of all representations: chop it up and every irreducible falls out, each as many times as its dimension. The counting identity is remarkably constraining:

Example 4.16. (i) *Cyclic groups, completeness at last.* The n representations $\chi_0, \ldots, \chi_{n-1}$ of C_n from Lecture 3 satisfy $\sum 1^2 = n = |C_n|$: they exhaust the irreducible representations of C_n, as promised. (ii) S_3 *is finished too.* The trivial, sign and standard representations give $1^2 + 1^2 + 2^2 = 6 = |S_3|$: the list found in Lecture 3 is complete, with no search required. (iii) *Constraint solving.* For any nonabelian group of order 8, the possible dimension patterns $(n_1, n_2, \ldots)$ with $\sum n_i^2 = 8$, $n_1 = 1$ (the trivial representation always exists) and not all $n_i = 1$ (that would force abelian, by Corollary 4.22 below) leave exactly one option:

$$8 = 1 + 1 + 1 + 1 + 4 :$$

four one-dimensional representations and one two-dimensional. Both nonabelian groups of order 8 — the dihedral group D_4 and the quaternion group Q_8 of Sheet 4 — are forced into this same pattern before we compute anything.

4.4 Counting the irreducibles: conjugacy classes

How many irreducible representations does a finite group have? The answer ties the representation theory back to the conjugacy structure of Lecture 2, and is the second pillar (with $\sum n_i^2 = |G|$) on which the character tables of Lecture 5 stand.

Theorem 4.17. *The number of inequivalent complex irreducible representations of a finite group equals the number of its conjugacy classes.*

The proof identifies both numbers with the dimension of one and the same space: the *centre* of the group algebra,

$$Z\big(\mathbb{C}[G]\big) = \big\{\, z \in \mathbb{C}[G] \;:\; za = az \text{ for all } a \in \mathbb{C}[G] \,\big\}.$$

That the class count appears there is elementary and we prove it at once; that the representation count does is the starred material below.

Lemma 4.18 (Class sums). *For each conjugacy class $C \subseteq G$ let $z_C = \sum_{g\in C} e_g \in \mathbb{C}[G]$. The class sums form a basis of $Z(\mathbb{C}[G])$; in particular*

$$\dim Z(\mathbb{C}[G]) = \#\{\text{conjugacy classes of } G\}.$$

Proof. An element $a = \sum_g a_g e_g$ is central iff it commutes with every basis element, i.e. $e_h\, a\, e_h^{-1} = a$ for all h. The left side is $\sum_g a_g e_{hgh^{-1}} = \sum_g a_{h^{-1}gh}\, e_g$, so centrality says exactly that the coefficient function is constant on conjugacy classes: $a_{hgh^{-1}} = a_g$ for all g, h. Such elements are precisely the linear combinations of class sums, and the z_C are linearly independent because distinct classes are disjoint. ∎

Structure of the remaining argument. Extending each irreducible representation linearly to the algebra gives a map

$$\Phi\colon\ \mathbb{C}[G] \longrightarrow \bigoplus_i \operatorname{End}(V^{(i)}), \qquad \Phi(a) = \big(\rho^{(1)}(a),\, \rho^{(2)}(a), \dots\big),$$

into the direct sum of full matrix algebras. One shows Φ is an *isomorphism of algebras* — injective because the regular representation is faithful and built from the $V^{(i)}$, surjective by the dimension count $\sum n_i^2 = |G|$ of Theorem 4.15. The centre of a direct sum of matrix algebras is easy to compute: each block contributes only scalar matrices (Schur again, in algebraic clothing), one independent scalar per block. Hence $\dim Z(\mathbb{C}[G])$ equals the number of blocks, i.e. of irreducible representations, and Lemma 4.18 finishes the proof. The details follow, starred; the statement of Theorem 4.17 is what the rest of the course uses.

⋆ The group algebra is a sum of matrix algebras

Theorem 4.19 (⋆). *The map Φ above is an isomorphism of algebras:*

$$\mathbb{C}[G] \;\cong\; \bigoplus_i \operatorname{End}(V^{(i)}) \;\cong\; \bigoplus_i M_{n_i}(\mathbb{C}).$$

Proof (⋆). Φ is an algebra homomorphism since each extended $\rho^{(i)}$ is (linearity, plus multiplicativity on the basis).

Injectivity. Suppose $\Phi(a) = 0$, i.e. a acts as zero in every irreducible representation. By Maschke's theorem every representation is a direct sum of irreducibles, so a acts as zero in *every* representation of G — in particular in the regular representation. But the regular representation detects every algebra element: $\rho^{\mathrm{reg}}(a)\, e_e = a$. Hence $a = 0$.

Surjectivity. Both sides have the same finite dimension, $\dim \mathbb{C}[G] = |G| = \sum_i n_i^2 = \sum_i \dim \operatorname{End}(V^{(i)})$ by Theorem 4.15; an injective linear map between spaces of equal finite dimension is surjective. ∎

Lemma 4.20 (⋆). *The centre of the matrix algebra $\operatorname{End}(V) \cong M_n(\mathbb{C})$ consists of the scalar matrices: $Z(M_n(\mathbb{C})) = \mathbb{C}\mathbb{1}$.*

Proof (⋆). Let Z commute with all matrices, in particular with the matrix units $E^{\mu\nu}$. Comparing entries of $ZE^{\mu\nu} = E^{\mu\nu}Z$ gives $Z_{\alpha\mu}\delta_{\nu\beta} = \delta_{\alpha\mu}Z_{\nu\beta}$ for all indices; setting $\alpha = \mu$, $\beta = \nu$ yields $Z_{\mu\mu} = Z_{\nu\nu}$, and setting $\beta = \nu$, $\alpha \neq \mu$ yields $Z_{\alpha\mu} = 0$ off the diagonal. ∎

Proof of Theorem 4.17 (⋆). An element of the centre of a direct sum of algebras is a tuple of central elements of the summands, so by Theorem 4.19 and Lemma 4.20,

$$Z(\mathbb{C}[G]) \;\cong\; \bigoplus_i Z(\operatorname{End}(V^{(i)})) = \bigoplus_i \mathbb{C}\mathbb{1}_{V^{(i)}},$$

of dimension equal to the number of irreducible representations. By Lemma 4.18 the same dimension is the number of conjugacy classes. ■

Theorem 4.19 deserves a comment beyond its role here: it says the group algebra of a finite group is, as an algebra, nothing but a direct sum of full matrix algebras, with block sizes the dimensions of the irreducible representations. The great orthogonality theorem is this statement seen through the inner product; the character theory of Lecture 5 is this statement seen through the trace.

With both counting theorems available, representation theory becomes arithmetic:

Example 4.21 (Dimension bookkeeping)**.**

G	#classes	$\lvert G\rvert$	dimensions (n_i)
C_n	n	n	$(1, 1, \dots, 1)$
$S_3 \cong D_3$	3	6	$(1, 1, 2)$
D_4	5	8	$(1, 1, 1, 1, 2)$
Q_8	5	8	$(1, 1, 1, 1, 2)$
D_5	4	10	$(1, 1, 2, 2)$

In each line the dimensions are *forced*: they are the unique multiset of positive integers, one of them 1 (the trivial representation), of the right cardinality, with squares summing to $|G|$. The class counts come from Lecture 2 (D_4 and D_5 were computed there; Q_8 is on Sheet 4). Note the cautionary tale in the table: D_4 and Q_8 have identical dimension data, yet are not isomorphic — representation dimensions do not determine the group.

Corollary 4.22. *A finite group is abelian if and only if all of its irreducible representations are one-dimensional, and in that case it has exactly $|G|$ of them.*

Proof. If G is abelian, Corollary 4.5 gives $n_i = 1$ for all i, and then $\sum_i n_i^2 = |G|$ forces $|G|$ of them. Conversely, if all $n_i = 1$, the number of irreducibles is $\sum n_i^2 = |G|$, which by Theorem 4.17 is the number of conjugacy classes; so every class is a singleton, i.e. $hgh^{-1} = g$ for all g, h: abelian. ■

4.5 Fourier analysis on a finite group

The two counting theorems upgrade the orthogonality of matrix elements into a completeness statement.

Theorem 4.23 (Completeness of matrix elements)**.** *The functions $\sqrt{n_i}\, D^{(i)}_{\mu\nu}$ (i over the irreducible representations, $1 \le \mu, \nu \le n_i$, matrices unitary) form an orthonormal* basis *of the space of all functions $G \to \mathbb{C}$ with the inner product $\langle \cdot, \cdot \rangle_G$. Every function ψ on G expands uniquely as*

$$\psi(g) = \sum_i \sum_{\mu,\nu=1}^{n_i} c^{(i)}_{\mu\nu}\, D^{(i)}_{\mu\nu}(g), \qquad c^{(i)}_{\mu\nu} = n_i \left\langle D^{(i)}_{\mu\nu}, \psi \right\rangle_G .$$

Proof. Orthonormality is Theorem 4.9; the count $\sum_i n_i^2 = |G|$ (Theorem 4.15) says the set has exactly the dimension of the function space, so it is a basis. The coefficient formula is the standard orthonormal expansion. ■

For $G = C_n$ the matrix elements are the characters $\chi_k(r^m) = e^{2\pi i k m/n}$, and the theorem *is* the discrete Fourier transform: expansion of an arbitrary function of m in exponential modes, with the familiar coefficient integrals replaced by averages over the group. Everything in this lecture is, from this angle, the generalisation of Fourier analysis to noncommutative groups — the "frequencies" becoming matrix-valued. The same idea, pushed from finite groups to compact Lie groups in Part II, will exhibit the spherical harmonics as matrix elements of the rotation group, with the Peter–Weyl theorem in the role of Theorem 4.23.

Remark 4.24 (Computer algebra)**.** The great orthogonality theorem is ideally suited to numerical spot-checks. For D_3, tabulate the six functions of Example 4.11 as vectors of values over the six group elements, rescale by $\sqrt{n_i}$, and verify that their Gram matrix with respect to $\langle\cdot,\cdot\rangle_G$ is the 6×6 identity — six lines of Mathematica. Sheet 4, Exercise 2 checks two representative matrix-element orthogonality identities for these same D_3 matrices; it does not assign a full Gram-matrix computation. As an optional, unassigned computer-algebra check, repeat the full calculation for D_4, whose two-dimensional representation is generated by $D(r) = R(\pi/2)$ and $D(s) = \mathrm{diag}(-1, 1)$ and whose four one-dimensional representations are fixed by the possible signs $(\nu(r), \nu(s)) = (\pm 1, \pm 1)$; there the Gram matrix is the 8×8 identity. Exact arithmetic (`Cos`, `Sin` of exact angles, `FullSimplify`) keeps the verification symbolic rather than floating-point.

Summary.

- *Schur's lemma.* An intertwiner of irreducibles is zero or an isomorphism (Theorem 4.2); over $\mathbb{C}$ it is a scalar (Theorem 4.3), so abelian groups have one-dimensional irreducibles (Corollary 4.5).
- *Great orthogonality theorem.* Matrix elements of the irreducibles are mutually orthogonal, with norm fixed by the dimension (Theorem 4.9).
- *Sum of squares.* $\sum_i n_i^2 = |G|$, from decomposing the regular representation (Theorem 4.15).
- *Counting.* The number of irreducibles equals the number of conjugacy classes (Theorem 4.17).
- *Completeness.* The matrix elements of the irreducibles form a basis of the functions on G (Theorem 4.23).

Tutorial. Exercise Sheet 4.

References

Serre, *Linear Representations of Finite Groups*, Pt. I (Schur's lemma and the orthogonality relations); Jones Pt. 4; Tung Ch. 3; Fulton–Harris §2.

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The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

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